

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/333035666>

Two new records of *Dendronephthya* octocorals (Family: Nephtheidae) from Andaman and Nicobar Islands, India

Article in *Indian Journal of Geo-Marine Sciences* · March 2019

CITATIONS

4

READS

419

4 authors, including:



J S Yogesh Kumar

Zoological Survey of India

74 PUBLICATIONS 258 CITATIONS

[SEE PROFILE](#)



C. Raghunathan

Zoological Survey of India

433 PUBLICATIONS 1,059 CITATIONS

[SEE PROFILE](#)

Two new records of *Dendronephthya* octocorals (Family: Nephtheidae) from Andaman and Nicobar Islands, India

J.S. Yogesh Kumar^{1*}, S. Geetha², C. Raghunathan³, & R. Sornaraj²

¹ Marine Aquarium and Regional Centre, Zoological Survey of India, (Ministry of Environment, Forest and Climate Change), Government of India, Digha, West Bengal, India

² Research Department of Zoology, Kamaraj College (Manonmaniam Sundaranar University), Thoothukudi, Tamil Nadu, India

³ Zoological Survey of India, (Ministry of Environment, Forest and Climate Change), Government of India, M Block, New Alipore, Kolkata, India

*[E-mail: coralyogesh@yahoo.co.in]

Received 04 August 2017; revised 23 November 2017

An extensive survey to explore the variety and distribution of octocorals and associated faunal community in and around the Andaman and Nicobar Islands yielded two species (*Dendronephthya mucronata* and *D. savignyi*), which is new zoogeographical record in India. The elaborate description, distribution and morphological characters are presented in this paper. The literature reveals that so far 55 species of *Dendronephthya* octocorals have been recorded from India.

[**Keywords:** Soft corals, Octocorals, *Dendronephthya*, Carnation corals, Andaman and Nicobar islands]

Introduction

The genus *Dendronephthya* (Cnidaria: Octocorallia: Alcyonacea: Nephtheidae) is an azooxanthellate and colourful soft coral reported in the warm coastal waters of the Indo-Pacific region, supporting reef formation of coral reef ecosystems throughout the world. It is commonly called carnation coral in the aquarium trade. The colours are crazy bright red, purple, pink, green, white orange, yellow, and mixes of these as well. Kukenthal¹ described this genus, dividing it into three main divisions: Glomerate, Divaricate and Umbellate, based on the colonial growth forms and separated this genus from the genera of *Spongodes*². Later, Tixier-Durivault and Prevorsek³ divided the genus *Dendronephthya* into *Spongodes*, *Roxasia* and *Morchellana*, based on species in the Glomerate, Divaricate and Umbellate divisions and followed the classification of Kukenthal. Utinomi⁴ created three subgenera: *Dendronephthya*, *Roxasia* and *Morchellana*. Thereafter, Verseveldt⁵⁻¹⁰ and Imahara¹¹⁻¹⁶ used the subgenera of autonomy in some reports.

Taxonomic studies carried out by various researchers described 255 species till today¹⁷⁻²⁸. In India, studies on the *Dendronephthya* octocoral taxonomy and distribution are very much

circumscribed to the works of Henderson¹⁷. The above studies reported 53 new species. Henderson's work was based on the material made available from the Royal Indian Marine Survey ship collections from several Indian Ocean locations, of which 34 species were reported from Andaman and Nicobar islands. These circumstances warrant a comprehensive study on the systematic and distribution of the Indian *Dendronephthya*. In the present study, two *Dendronephthya* species are described and an updated checklist is also provided for further studies in India.

Materials and Methods

Andaman and Nicobar archipelago has 572 reef islands between Myanmar (North) to Sumatra (South) 6° to 14° North and 92° to 94° East with 2000 km² coral reef cover²⁹. The *Dendronephthya* octocoral taxonomical study was conducted at Andaman and Nicobar islands during the period from June 2013 to August 2015. The samples were collected in the study area by SCUBA diving up to 40 m depth. The specimens were identified based on the morphological characterises, such as growth form, colour, size of each part of the colony, polyp, and sclerite. The glosses of each piece were taped and live colonies photo graphed using underwater digital camera (Sony

T900 and Canon G10). Sclerites were extracted using 5% sodium hypochlorite³⁰.

Results

A total of 55 species of *Dendronephthya* octocorals have been recorded from India (Table 1), of which two species are new to India. Till today, 40 species are reported from Andaman and Nicobar islands, seven species from the east coast (Ganjam coast, Orissa), four species from the Gulf of Kachchh, two species from the Gulf of Mannar, two species from Lakshadweep islands and three samples identified up to genus level from the west coast (Kerala coast). The species identification of the present study was made based on variations in morphological features of sclerites.

Systematic accounts

Phylum : Cnidaria Hatschek³¹
 Class : Anthozoa Ehrenberg³²
 Subclass : Octocorallia Haeckel³³
 Order : Alcyonacea Lamouroux³⁴
 Family : Nephtheidae Gray³⁵
 Genus : *Dendronephthya* Kukenthal¹

1. *Dendronephthya mucronata*³⁶ (Fig. 1)

Material Examined: ZSI/ANRC-7703 Sample, 13.5 cm long and 9.8 cm broad and 5-10 m deep; Pongibalu (Latitude 11°31.030 N; Longitude 092°39.159 E), South Andaman; shallow to reef edge with turbid water.

Description: Colonies under glomerate type; yellowish white stalk and branches, orange-red colour in stem and polyps, the stalk or trunk is thick measuring 3.0 cm, about one-third of the total height of the colony. The polyps are arranged in groups of four to eight. The sclerites are arranged in eight double rows of straight or curved spindles of 0.3–6.5 mm size; in stalk sclerites are short, thick spindles, clubs, three and four rayed and oval forms; in the stem spindles up to 4.0 mm in size.

Distribution: Indo Pacific; India: Andaman and Nicobar islands.

2. *Dendronephthya savignyi*³² (Fig. 2)

Material Examined: ZSI/ANRC-7702 Sample, 8.6 cm wide and 4.2 cm high and 10.0 m deep; South Button Island (Latitude 12°13.4 N; Longitude 092°01.334 E), South Andaman; shallow to reef edge with turbid water.

Description: Colonies under umbellate type with white colour in stalk, stem, and branches, whereas

polyp is greyish white. The stalks are very short and polyps are arranged in groups of three to eight and may occur in small groups on flattened branches. Sclerites are flattened curved spindles (0.24 mm); in stem spindles either curved or straight (1.28–2.4 mm); in stalk sclerites are spindles curved, rods, three rayed and four rays (0.25–1.5 mm).

Distribution: Indo-Pacific; India: Andaman and Nicobar islands.

Discussion

Octocorals are distributed worldwide and are important component of coral reef ecosystems in the Indo-Pacific region³⁷. Taxonomical studies were carried out by different researchers from different parts of the world, such as Southern Vietnam³⁸, the Red Sea^{39, 40}, the southern Red Sea⁴¹, eastern and southern Africa⁴²⁻⁴⁴, the Seychelles⁴⁵, American Samoa⁴⁶, Southwestern Australia⁴⁷, Southern Taiwan⁴⁰, the South China Sea, the Indo-Pacific Ocean⁴⁸, and Indonesia⁴⁹.

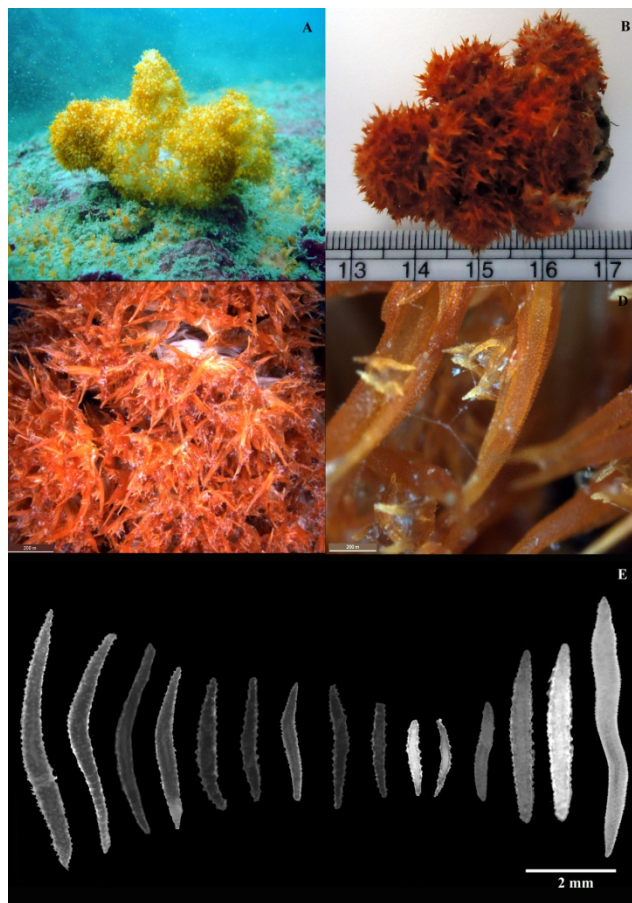


Fig. 1 — A: Live underwater picture of *Dendronephthya mucronata*, B: Deposit specimen ZSI/ANRC, C & D: Polyp arrangement, E: Spindle sclerites in different size

Table 1 — Checklist of Genus *Dendronephthya* sp. (Octocorallia) from India

S. No.	List of Species from India	AN	LAK	GOM	GOK	EC	WC	Remarks
1.	<i>Dendronephthya albegilva</i>	*	-	-	-	-	-	-
2.	<i>Dendronephthya andamanensis</i>	*	-	-	-	-	-	-
3.	<i>Dendronephthya arbuscula</i>	*	-	-	-	-	-	-
4.	<i>Dendronephthya aurora</i>	-	-	*	-	-	-	-
5.	<i>Dendronephthya biformata</i>	-	-	-	-	*	-	-
6.	<i>Dendronephthya booleyi</i>	*	-	-	-	-	-	-
7.	<i>Dendronephthya brachycaulos</i>	*	-	-	-	-	-	-
8.	<i>Dendronephthya brevirama</i>	*	*	-	*	-	-	-
9.	<i>Dendronephthya cervicornus</i>	*	-	-	-	-	-	-
10.	<i>Dendronephthya cocosiensis</i>	*	-	-	-	-	-	-
11.	<i>Dendronephthya conica</i>	*	-	-	-	-	-	-
12.	<i>Dendronephthya costatorubra</i>	*	-	-	-	-	-	-
13.	<i>Dendronephthya delicatissima</i>	*	-	-	-	-	-	-
14.	<i>Dendronephthya dendrophyta</i>	-	-	-	*	-	-	-
15.	<i>Dendronephthya dichotoma</i>	-	-	-	-	*	-	-
16.	<i>Dendronephthya divaricata</i>	*	-	-	-	-	-	-
17.	<i>Dendronephthya elegans</i>	*	-	-	-	-	-	-
18.	<i>Dendronephthya elongate</i>	*	-	-	--	-	-	-
19.	<i>Dendronephthya foliata</i>	*	-	-	-	-	-	-
20.	<i>Dendronephthya ganjamensis</i>	-	-	-	-	*	-	-
21.	<i>Dendronephthya gilva</i>	*	-	-	-	-	-	-
22.	<i>Dendronephthya harrisoni</i>	*	-	-	-	--	-	-
23.	<i>Dendronephthya hemprichi</i>	-	-	*	-	-	-	-
24.	<i>Dendronephthya irregularis</i>	-	-	-	*	-	-	-
25.	<i>Dendronephthya koellikeri</i>	*	-	-	-	-	-	-
26.	<i>Dendronephthya lanxifera</i>	*	-	-	-	-	-	-
27.	<i>Dendronephthya longispina</i>	*	-	-	-	-	-	-
28.	<i>Dendronephthya macrocaulis</i>	*	-	-	-	-	-	-
29.	<i>Dendronephthya masoni</i>	*	-	-	-	-	-	-
30.	<i>Dendronephthya micracanthus</i>	-	*	-	*	-	-	-
31.	<i>Dendronephthya microspiculata</i>	*	-	-	-	-	-	-
32.	<i>Dendronephthya mirabilis</i>	*	-	-	-	-	-	-
33.	<i>Dendronephthya mucronata</i>	*	-	-	-	-	-	New to India
34.	<i>Dendronephthya multispinosa</i>	*	-	-	-	-	-	-
35.	<i>Dendronephthya nicobarensis</i>	*	-	-	-	-	-	-
36.	<i>Dendronephthya obtuse</i>	*	-	-	-	-	-	-
37.	<i>Dendronephthya ochracea</i>	*	-	-	-	-	-	-
38.	<i>Dendronephthya orientalis</i>	*	-	-	-	-	-	-
39.	<i>Dendronephthya ovate</i>	-	-	-	-	*	-	-
40.	<i>Dendronephthya pallid</i>	*	-	-	-	-	-	-
41.	<i>Dendronephthya ellucid</i>	*	-	-	-	-	-	-
42.	<i>Dendronephthya pentagona</i>	*	-	-	-	-	-	-
43.	<i>Dendronephthya purpurea</i>	*	-	-	-	-	-	-
44.	<i>Dendronephthya quadrata</i>	*	-	-	-	-	-	-
45.	<i>Dendronephthya rubeola</i>	*	-	-	-	-	-	-
46.	<i>Dendronephthya rubescens</i>	-	-	-	-	*	-	-
47.	<i>Dendronephthya savignyi</i>	*	-	-	-	-	-	New to India
48.	<i>Dendronephthya thuja</i>	-	-	-	-	*	-	-
49.	<i>Dendronephthya thompsoni</i>	-	-	-	-	*	-	-
50.	<i>Dendronephthya uliginosa</i>	*	-	-	-	-	-	-
51.	<i>Dendronephthya variata</i>	*	-	-	-	-	-	-
52.	<i>Dendronephthya varicolor</i>	*	-	-	-	-	-	-
53.	<i>Dendronephthya</i> sp.1	-	-	-	-	-	*	-
54.	<i>Dendronephthya</i> sp.2	-	-	-	-	-	*	-
55.	<i>Dendronephthya</i> sp.3	-	-	-	-	-	*	-
Total		40	2	2	4	7	1	2

* Present; AN - Andaman and Nicobar; LAK – Lakshadweep; GOM – Gulf of Mannar; GOK – Gulf of Kachchh; EC – East Coast; WC – West Coast

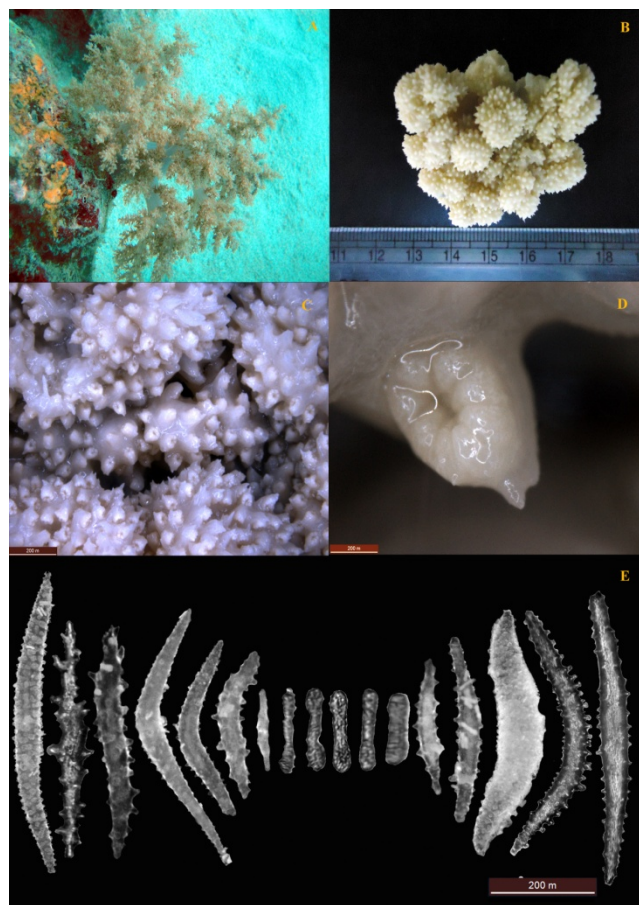


Fig. 2 — A: Live underwater picture of *Dendronephthya savignyi*, B: Deposited specimen ZSI/ANRC, C&D: Polyp arrangement, E: Spindle sclerites in different size

The genus *Dendronephthya* is divided into three groups (Glomeratae, Divaricatae and Umbellatae) based on their morphology¹. Henderson reported 67 species of *Dendronephthya* from The Indian Ocean, among which 58 species were newly described under this genus¹⁷. After 1909, the present study yielded two species, *Dendronephthya mucronata* and *D. savignyi* from Andaman Sea. Also, these two species were reported from German Museum collection by Jurgens and Reinicke⁵⁰ and many other species under this genus were reported from India^{17, 18, 23, 3, 51–57}.

Octocorals diversity is reported to be rich in India; however, efforts to study the distribution, abundance, species richness, species diversity and ecology of octocorals are very limited. The genus *Dendronephthya* mostly tends to occur in deeper region and strong water current. This may be one of the reasons for the study and collection of samples. Andaman and Nicobar islands are very rich in

biodiversity and more investigation of the population and taxonomy is needed.

Conclusion

The present updated list provides 55 species of *Dendronephthya* from Indian waters based on literature compilation. In addition to this, *Dendronephthya mucronata* and *Dendronephthya savignyi* were reported in the present study. It is recommended that extensive studies are needed in the future.

Acknowledgements

Authors are grateful to the Director, Zoological Survey of India, Kolkata and thankful to the Department of Science and Technology (DST) and Science and Engineering Research Board (SERB) for financial support to undertake the survey and Ministry of Environment of Forest and Climate Change (MoEFCC) for the permission granted to survey the Islands.

References

- 1 Kükenthal, W., Versuch einer revision der Alcyonarien. II. Die familie der Nephthyiden. 2. Teil. Die gattungen *Dendronephthya* n.g. und *Stereonephthya* n.g. *Zoologische Jahrbücher (Systematik)*, 21 (1905) pp. 503-726 (in German).
- 2 Lesson, R. P., Illustrations de zoologie ou recueil de figures d'animaux peintes d'après nature, *Arthus Bertrand, Paris*, 23 (1831) pp.1-354.
- 3 Tixier-Durivault A, Prevorsek M., Révision de la famille des Nephtheidae. 1. Le genre *Spongodes* Lesson 1831. *Mémoires du Muséum National d'Histoire Naturelle (Nouvelle Série), Série A, Zoologie*, 20 (1959) pp. 1-151(in French).
- 4 Utinomi, H., *Preliminary list of octocorals of Sagami Bay deposited in the Biological Laboratory of the Imperial Household*. Publications of the Seto Marine Biological Laboratory, 10 (1962) pp. 105-108.
- 5 Verseveldt, J., Biological results of the Snellius Expedition XXII. Octocorallia from the Malay Archipelago (Part II). *Zoologische Verhandelingen*, 80 (1966) pp. 1-107.
- 6 Verseveldt, J., *Preliminary note on some new Octocorallia from Madagascar*. Proceedings Koninklijke Nederlandse Akademie van Wetenschappen Amsterdam (Series C), 71 (1968) pp. 52-59.
- 7 Verseveldt, J., Report on some Octocorallia(Alcyonacea) from the northern Red Sea. *Israel Journal of Zoology*, 19 (1970) pp. 209-229.
- 8 Verseveldt, J., *Octocorallia from north-western Madagascar* (Parts IIIA, IIIB). Proceedings Koninklijke Nederlandse Akademie van Wetenschappen Amsterdam (Series C), 76 (1973) pp. 69-100.
- 9 Verseveldt, J., Alcyonacea (Octocorallia) from the Red Sea, with a discussion of a new *Sinularia* species from Ceylon. *Israel Journal of Zoology*, 23 (1974) pp.1-37.

- 10 Verseveldt, J., Australian Octocorallia (Coelenterata). *Australian Journal of Marine and Freshwater Research*, 28 (1977) pp. 171-240.
- 11 Imahara, Y., Nephtheid octocorals from Suruga Bay, the Pacific coast of central Japan. *Annotationes Zoologicae Japonenses*, 50 (1977) pp. 164-173.
- 12 Imahara, Y., Report on the Octocorallia from the Ryukyu Islands of Japan. *Bulletin of Institute Oceanic Research and Development, Tokai University*, 11/12 (1991) pp. 59-94.
- 13 Imahara, Y., Catalogue of the invertebrate materials in the Wakayama Prefectural Museum of Natural History, 1(1982 - 1992). *Bulletin of the Wakayama Prefectural Museum of Natural History*, 14 (1996a) pp. 41-86. (in Japanese with English abstract).
- 14 Imahara, Y., *Previously recorded octocorals from Japan and adjacent seas*. Precious Corals & Octocoral Research, 45 (1996b) pp. 17-44.
- 15 Imahara, Y., *Octocorals of the Döderlein collection*. Rep Activ 2000-2002. Grants-in-Aid for Scientific Research(B) (No. 12575008). Japan Society for the Promotion of Science, Tokyo, (2003a) pp. 259-266.
- 16 Imahara, Y., *Preliminary list of Japanese octocorals deposited in the various museum in Europe, excluding Döderlein's collection*. Rep Activ 2000-2002. Grants-in-Aid for Scientific Research (B) (No. 12575008). Japan Society for the Promotion of Science, Tokyo (2003b) pp. 267-289.
- 17 Henderson, W. D., *An account of the alcyonarians collected by the Royal Indian Marine Survey Ship Investigator in the Indian Ocean*, with a report on the species of Dendronephthya by W.D. Henderson. II. The alcyonarians of the littoral area. The Indian Museum, Calcutta, (1909) pp. 1-319.
- 18 Thomson, J. A. and Simpson, J. J., *An account of the Alcyonarians collected by the RIM SS Investigator in the Indian Ocean*. II. The Alcyonarians of the littoral area: XII. 1909 pp. 1-319.
- 19 Nutting, C. C., *Descriptions of the Alcyonaria collected by the U.S. Fisheries steamer "Albatross", mainly in Japanese waters, during 1906*. Proceedings of the United States National Museum, 43 (1912) pp. 1-104.
- 20 Sherrioffs, W. R., *Evolution within the genus Dendronephthya (Spongodes) (Alcyonaria), with descriptions of a number of species*. Proceedings of the Zoological Society of London, 92 (1922) pp. 33-77.
- 21 Thomson, J. A., and Dean, L. M. I., *The Alcyonacea of the Siboga Expedition with an addendum to the Gorgonacea*. Siboga Expedition Monographs, Leiden, 13d (1931) pp. 1-227.
- 22 Utinomi, H., *Dendronephthya of Japan. I. Dendronephthya collected chiefly along the coast of Kii Peninsula*. Publications of the Seto Marine Biological Laboratory, 2 (1952) pp. 161-212.
- 23 Song, J. I., A study on the classification of the Korean Anthozoa. 2. Alcyonacea. *Korean Journal of Zoology*, 19 (1976) pp. 51-62.
- 24 Utinomi, H., *Dendronephthya of Japan. II. New species and new records of Dendronephthya and the allied Stereonephthya from Kii region*. Publications of the Seto Marine Biological Laboratory, 3 (1954) pp. 319-338.
- 25 Rho, B. J. and Song, J. I., A study on the classification of the Korean Anthozoa 3. Alcyonacea and Pennatulacea. *Journal of the Korean Research Institute for Better Living*, 19 (1977) pp. 21.
- 26 Li, G. Q., Deng Z. W., Guan H. S., Ofwegen L. P. van, Proksch P. and Lin, W. H., Steroids from the soft coral Dendronephthya sp.. *Steroids* 70 (2005) pp. 13-18.
- 27 Ma, A., Deng, Z., Ofwegen, L. P., van, Bayer, M., Proksch, P., Lin, W. 2008. Dendronpholides A-R, cembranoid diterpenes from the Chinese soft coral Dendronephthya sp.. *Journal of Natural Products* 71 (2008) pp. 1152-1160.
- 28 Ma, A., Deng Z., Ofwegen, L. P. van, Bayer, M., Proksch, P. and Lin, W., Dendronpholides A-R, Cembranoid Diterpenes from the Chinese Soft Coral Dendronephthya sp. (vol 71, pg 1133, 2008). *Journal of Natural Products* 73 (2010) pp. 1026-1026.
- 29 Turner, J. R., Vousden, D., Klaus, R., Satyanarayana, C., Fenner, D., Venkataraman, K., Rajan, P. T. and Subba Rao, N.V., *Report of Phase I: Remote Sensing and Rapid Site Assessment Survey*, April 2001. Coral Reef Systems of the Andaman Islands. Government of India, UNDP & GEF (2001).
- 30 Bayer, F. M. *The Shallow water octocorallia of the West Indian region*. Stud. Fauna Curacao, 12 (1961) pp. 1-373.
- 31 Hatschek, *Lehrbuch der Zoologie* G. Fischer, Jena (1888) pp. 432.
- 32 Ehrenberg, C. G., *Die Korallenthiere des rothen Meeres*, Berlin, (1834) P. 156.
- 33 Haeckel, E., *Generelle Morphologie der Organismen*. G. Reimer, Berlin, (1866) 1036 pp.
- 34 Lamouroux, *Histoire des Polypiers coralligenes flexible*. Caen (1816).
- 35 Gray, J. E., *Description of some new species of Spogcodes and of a new allied genus (Morchellana) in the collection of the British Museum*. Proceeding of the Zoological Society of London, 30 (1862) pp. 27-31.
- 36 Pütter, A., *Alcyonaceen des Breslauer Museums*. Zoologische Jahrbücher (Systematik), 13 (1900) pp. 443-462 (in German).
- 37 Fabricius, K., and Alderslade, P., *Soft corals and sea fans: a comprehensive guide to the tropical shallow water genera of the central-west Pacific, the Indian Ocean and the Red Sea*. Australian Institute of Marine Science (AIMS), (2001) Jun 1.
- 38 Malyutin, A. N., Two new species of Sinularia (Octocorallia: Alcyonacea) from South Vietnam. *Asian Mar. Biol.* 7 (1990) pp. 9-14.
- 39 Perkol-Finkel, S. And Benayahu, Y., Community structure of stony and soft corals on vertical unplanned artificial reefs in Eilat (Red Sea): Comparison to natural reefs. *Coral reefs* (2004) pp.122.
- 40 Benayahu, Y., M. -S. Jeng, S. Perkol-Finkel, C. and Dai, F., Soft corals (Octocorallia, Alcyonacea) from Southern Taiwan. II. Species diversity and distributional patterns. *Zool. Studies*, 43 (2004) pp. 548-560.
- 41 Benayahu, Y., Yosief, T., Schleyer, M. H., Soft corals (Octocorallia, Alcyonacea) of the Southern Red Sea. *Israel J. Zool.*, 48 (2002) pp. 273-283.
- 42 Williams, G.C., Two new genera of soft corals (Anthozoa: Alcyoniidae) from South Africa, with a discussion

- of diversity and endemism in the southern African octocorallian fauna. *Proc. Calif. Acad. Sci.*, 52 (6) (2000) pp. 65-75.
- 43 Williams, G.C., Capitulate taxa of the soft coral genus *Eleutherobia* (Octocorallia: Alcyoniidae) from Palau and South Africa: A new species and a new combination. *Zool. Verh. Leiden.*, 345 (2003) pp. 419-436.
 - 44 Benayahu, Y., A. Shlagman and Schleyer, M. H., Corals of the South-west Indian Ocean VI. *Alcyonacea* (Octocorallia) of Mozambique with a discussion on soft corals latitudinal distribution along south equatorial East African reefs. *Zoologische Verhandelingen Leiden*, 344 (2003) pp. 49-57.
 - 45 Malyutin, A. N., Octocorallia from the Seychelles Islands with some ecological observations. *Atoll Res. Bull.* 367 (1992) pp. 1-9.
 - 46 Cornish, A. S. And DiDonato E. M., Resurvey of a reef flat in American Samoa after 85 years reveals devastation to a soft coral (Alcyonacea) community. *Mar. Pollut. Bull.* 48 (2004) pp.768-777.
 - 47 Alderslade, P., A new genus and species of soft coral (Octocorallia: Alcyonacea: Alcyoniidae) from Lord Howe Island, Australia. *Zool. Verh. Leiden.*, 345 (2003) pp. 19-29.
 - 48 Ofwegen, L. P. V., A new genus of nephtheid soft corals (Octocorallia: Alcyonacea: Nephtheidae) from the IndoPacific. *Zool. Meded. Leiden* 79 (2005) pp. 1-236.
 - 49 Ofwegen, L. P. V., *Lobophytum jasparsi* spec. Nov. from Indonesia (Coelenterata: Octocorallia: Alcyonacea). *Zool. Med. Leiden* 73 (8) (1999) pp. 177-185.
 - 50 Jurgens, Lars and Reinicke, G. B., *The soft coral genus Dendronephthya Kukenthal, 1905 (Octocorallia): Inventory and investigations of the type material in German Museum Collections*. Organisms, Diversity, Evolution, Electr. Suppl. (2005) 13. (<http://www.senckenberg.de/odes/Juergens.pdf>)
 - 51 Tixier-Durivault, A. and Prevorsek, M., 1960. *Le genre Roxasia (Alcyonaria, Nephtheidae).— Spolia. Zool. Mus. Haun.* 18 (1960) pp.1-296, figs. 1-256.
 - 52 Tixier-Durivault, A. and M. Prevorsek, 1962. *Le genre Morchellana (Alcyonaria, Nephtheidae).— Spolia. Zool. Mus. Haun.* 19:1-239, figs. 1-150.
 - 53 Verseveldt and Ofwegen, V., Five new species of the genus *Dendronephthya* Kukenthal, 1905, (Octocorallia: Nephtheidae) from the Indian Ocean, *Zoologische Mededelingen*, 65 (1991) pp. 155-169.
 - 54 Grossowicz, M. and Benayahu, Y., Differential morphological features of two *Dendronephthya* soft coral species suggest differences in feeding niches. *Mar. Biodiv.*, 42 (2012) pp.65-72.
 - 55 Hwang, S. J. and Song, J. I., Four new records of *Dendronephthya* species (Octocorallia: Alcyonacea: Nephtheidae) from Korea, *Anim. Syst. Evol. Divers.*, 29 (2): (2013) pp. 160-174.
 - 56 Sivaleela, G., and Padmanaban., Studies on the Alcyonacean fauna of Gulf of Mannar. Records of Zoological Survey of India 115(1): (2015) pp. 21-30.
 - 57 Mary, A. G., Sluka, R. B., and Lazarus, S., Octocoral diversity and distribution on the south-west Indian coast, *Marine Biodiversity Records*, 4 (2011) pp. 1-11.